

Manipulating HIIT Variables

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Martin Buchheit and Paul Laursen

This is a particularly important chapter in the book. Its contents may address questions that many students, coaches, scientists, and practitioners are likely to have. To introduce the content, we begin with our own personal stories of the journey we've taken to understanding high-intensity interval training (HIIT) variable manipulation. Ultimately, we probably have had many of the same questions about HIIT that you do. But we came at ours from somewhat different angles.

Paul Laursen's Experience

Following in my father's footsteps, I began running early as a child, which progressed into the sport of triathlon as the sport began to grow. I was good as an endurance athlete, but not the best. After an unsuccessful time trying to make a living from the sport of triathlon in my twenties, I developed a passion to try to understand endurance performance better. This was a selfish endeavor that stemmed directly from the frustration I felt with being unable to reach a certain standard in the sport.

When I began my doctoral studies in Australia at the beginning of the year 2000, there was a burning question I had at the time. If HIIT appeared so effective as a training strategy, what was the best type to use? Phrased another way, what was the best manipulation of exercise intensity, bout duration, and

recovery that elicits the most favorable adaptation and subsequent performance? How can you get the most bang for your buck out of a training session?

I personally found the related study by Dr. Nigel Stepto and colleagues (55) to be fascinating. In this study, the authors took 20 well-trained cyclists and divided them into five groups that performed vastly different interval sessions. While there were trends suggesting certain programs were better than others, what amazed me at the time was how nearly all programs were effective at least somehow at enhancing 40 km time trial performance, or related parameters, as well as the individual responses.

I distinctly recall asking for the thoughts of my PhD supervisor, Dr. David Jenkins. He replied, "Well, Paul, you know what they say, 'When it comes to training, there's more than one way to skin a cat.'" That was the first I'd ever heard the phrase, but it stuck with me, and I've since used it often. The old English proverb, "There's more than one way to skin a cat," meaning there's more than one way to do something, can sometimes be heard mentioned in the context of training chatter. As will be shown in this chapter, the number of moving parts within a high-intensity interval training (HIIT) session suggests this proverb well and truly does apply.

And that's when I began my own personal journey trying to examine the concept. For my PhD, I wanted to compare the effects of different forms of interval

training on physiology and its subsequent performance response. Just like Stepto et al. (55) showed, I too found that there was indeed more than one way to skin a cat when it comes to HIIT and its associated physiological effects and performance outcomes (42).

How many ways you ask? We'll get to that. But before we do, we should hear from the man who has taught me much about skinning cats, Martin Buchheit. Around a similar time, Martin had related questions and was following his own journey toward discovering these.

Martin Buchheit's Experience

I was just 14 years old in the summer of September 1993 when I began my training in the Strasbourg talent center in the pursuit of becoming a professional handball player (which I never made). But I was highly motivated to train in the hopes of gaining fitness to make a good impression on the coach to be picked for the national junior handball team. At the time, I was used to spending my holidays in the French Alps, where my mother had a summer job.

In the Alps, I visited the guys at the local ski club. Given their good level of performance on the slopes, I expected their training to be well structured and that they might be eager to help me work on my leg strength in the gym. They allowed me to join their group of about 15 skiers, ranging in age from 12 to 18 yr. For my first strength session, I was able to lift about a third of what the others could squat, and they immediately started to take the piss out of me for wanting to become a pro handball player. I retorted enthusiastically, "I don't need the same quad strength as you guys, and I'll prove that to you tomorrow in the running session."

The next morning involved an HIIT session, consisting of 3×8 min blocks of 15 s/15 s runs over undulating rocky terrain. For me (and others) to better cope, they decided to drive the valley, so that the altitude wouldn't be an issue. This was a day I thought I might die. Even the 12-year-old girls outperformed me, and I felt like an idiot. I was destroyed. The skiers were flying, running everywhere, while I couldn't catch my breath. That was my first ever exposure to a 15 s/15 s short-interval session. From that day on, I knew I needed to understand better how the exercise format worked, and that I'd need to learn to train like these athletes in order to reach the professional standards needed for elite handball.

Fast forward to 1998 during my university studies, making my first steps into coaching, where I met Alain Quintallet, the strength and conditioning coach of the French national handball team, who shared my same interest in short HIIT. In fact, we had both independently come to appreciate through our own experimentation that prescribing short HIITs using only an exercise and rest interval duration (such as 15 s/15 s) resulted in highly variable exercise formats, and in turn, different physiological and performance responses. For example, in my 15 s/15 s short interval, I would advise my players to run 70 to 80 m over 40 m shuttles and have passive rest in between. Alain would have his players running straight lines outdoors more than 80 m, with an active jog in between repetitions. What option was best? Why would you use one over the other? Together, over many sleepless nights, we began experimenting and discussing the training intricacies of the different programming variables that we could use to better define HIIT formats, conceiving the best way to skin the cat. These dialogues led to the development of the 30-15 IFT in 2000 (12). From here, I met the great Veronique Billat at Evry University in the south of Paris. She indirectly introduced me to the Cosmed K4 device, which allowed me to gain great insight into the physiology of short intervals, such as the effect of varying exercise intensity and modality and the relevance of the 30-15 IFT (chapter 2).

How Many Ways Can You Skin a Cat?

From two different parts of the world, following two distinctly different failed pathways (both unable to reach the professional ranks in their respective sports), we both sought to answer similar questions. Whether it was how to train an elite cyclist on a bike, train a triathlete on the road, or to prepare a handball player to excel on the court, our goal was the same—what's the best way to skin the cat when it comes to training so that performance potential is maximized? Put another way, in the context of an athlete's training program puzzle, what's the best way to shape the puzzle piece—in terms of the HIIT duration, intensity, and format—to make it fit into the performance development puzzle, so that an athlete's future development and performance is optimized?

In 2007, we began our collaboration as young scientists at Edith Cowan University, in Perth, Aus-

tralia, funded by the generous collaboration of our colleague, Professor Kazunori (Ken) Nosaka (figure 4.1). Over the subsequent decade, we worked on manipulating as many variables around HIIT as we could. Along with countless others who share our passion in this area, this chapter describes our interpretation of what we and others have found.

Manipulation of Interval Training Variables

Based on our stories, and as illustrated (figure 4.2), we feel there are 12 primary variables we can manipulate within a given HIIT session to skin the cat (11). The intensity and duration of work and relief intervals are the key influencing factors (4, 17). Then, the number of intervals, the number of series, and between-series recovery durations and intensities account for the total work performed. Exercise modality (i.e., running versus cycling or rowing, straight line versus uphill, change of direction running and ground surface) has to date received limited scientific interest, but it is clear that it represents a key variable to consider when programming HIIT, especially for team and racket sport athletes. Finally, both the environment (temperature, humidity, and partial pressure of oxygen) and nutritional factors are additional aspects that

can be manipulated to affect the acute physiological response. The manipulation of each variable in isolation likely has a direct impact on the metabolic, cardiopulmonary, and neuromuscular responses (chapter 3). When more than one variable is manipulated simultaneously, responses are more difficult to predict, since the factors are interrelated.

With these 12 moving parts, we had our colleague with expertise in mathematics, Dr Andrea Zignoli (CeRiSM Research Centre, Italy), work out for us using factorial analysis that in theory there could be 1,492,992 different combinations of HIIT sessions that could be created, and toward infinity. While there might be nearly 1.5 million ways to skin the cat with our HIIT programming, most of these would have no physiological meaning, so we'd best be more specific and purposeful than that toward achieving our objectives. Surely we can take this near infinite number of potential combinations and take a more useful approach to our programming. If we go back to where we began in chapter 1 with our individual athlete, alongside appreciating the important context of traditional approaches (chapter 2), how do we best adjust these 12 moving parts to achieve some of the desired physiological responses shown in chapter 3? The purpose of this chapter is to show some of the primary effects of these moving parts, so we can learn best how to go about manipulating



Figure 4.1 Colleagues gathering in 2007 at Edith Cowan University, Perth, Australia, to begin research into sport application of HIIT. Pictured left to right: Martin Buchheit, Kazunori (Ken) Nosaka, Paul Laursen, and Marc Quod.

Photo provided by Paul Laursen.

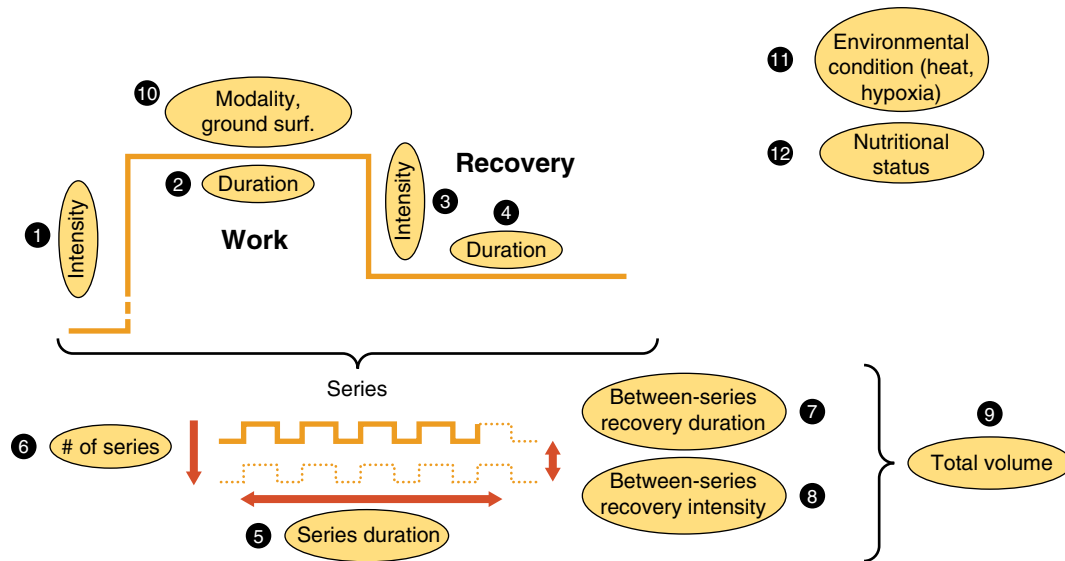


Figure 4.2 The 12 variables that can be manipulated to prescribe different HIIT sessions (11). These include (1) work bout intensity, (2) duration of the work bout, (3) recovery period intensity, (4) recovery period duration, (5) number of intervals or series duration, (6) number of interval bout series, and (7) the between-series recovery duration and (8) intensity. Variables 1 through 8 account for the (9) total work performed. Additionally, other factors that play a large role on the physiological outcome of an HIIT session are the (10) exercise mode and ground surface for run-based HIIT, (11) environment (heat and altitude), and (12) an athlete's nutrition practices.

a session's HIIT parameters to achieve our desired physiological targets.

Exercise Period Intensity

The first, and arguably most influential, characteristic of an HIIT session is the chosen intensity of the exercise period (figure 2.5). At the low end of the intensity spectrum is critical speed or maximal lactate steady state, an intensity largely maintainable for 30 to 60 min. At the high end, this intensity continuum consists of maximal sprinting speed or instantaneous peak power. This is the highest power the athlete can produce or fastest speed at which the athlete can move.

The determinants of all-out power or speed, from an absolute standpoint, relate to the phenotype of the individual in terms of his motor unit breakdown (slow- or fast-twitch muscle fiber predominance). While there is well-described variability of slow- and fast-twitch muscle proportions in human mixed muscle, the average human and athlete typically has about a 50% breakdown of each broad fiber type division (fast and slow twitch), acknowledging the fact that scientists have been able to take each of the broad-slow and fast-twitch categories and sub-

divide these even further. For argument's sake, and in the name of keeping things simple, we could say that we generally have slow (or type I), intermediate (or type IIa), and fast-twitch muscle fiber (or type IIb) groups. Additionally, the total numbers of motor units in a muscle group that can be engaged by the brain to produce the movement are influenced by the neuromuscular characteristics (chapter 3; (28)) and of course, training. Additionally, all-out maximal power or speed can relate to the individual's muscle mass or volume of motor units within a given functional muscle group, or cross-sectional area of muscle (39, 47).

When an individual motor unit of the millions in a working muscle group is stimulated by a nerve (called a motor neuron), it creates force and resulting movement until, bioenergetically or biochemically speaking, it fatigues. Fast-twitch motor units are fast to produce force, highly powerful, and predominantly glycolytic in nature; however, they fatigue rapidly. Alternatively, slow-twitch fibers produce less force slowly, but they are plentiful in mitochondria for ATP production and as a result are more resilient to fatigue. Intermediate muscle fibers are somewhere in between, having properties of both types. Throw an equal proportion of all these muscle

fibers into mixed human muscle, and this maximal power and speed ability creates an interesting relationship with our second crucial HIIT variable, the exercise period duration.

Exercise Period Duration

The exercise period duration of an HIIT bout is clearly dependent on the exercise intensity, which, as we just learned, is reliant on the proportion of muscle fibers engaged, along with their varying twitch characteristics, and the absolute amount (volume or size) of these. The maximum period over which an exercise task can be sustained at a constant speed or power output is an inverse and curvilinear function of the exercise intensity, and is specific to the muscle fiber (or volume/proportion of engaged muscle fibers). The higher the exercise intensity, the shorter its duration, and conversely, within one's own individual limits. Thus, the power/speed time continuum describes the interplay and trade-off of exercise intensity and duration in a particular individual completing a specific movement task or exercise.

When we perform exercise at our maximal intensity for varying times, and we plot the power and or speed versus that duration, we see a classic power law distribution profile. This helps provide us with some relevant mathematical constructs and terms we can use to help us understand training and performance. One is called the critical power, whereas the other is termed the anaerobic work capacity, often termed W' (W prime).

Critical Power (CP)

The critical power (CP) model describes the capacity to *sustain* a particular work rate as a function of time (t) for an individual. As first suggested by Hill in 1925 (38), when he plotted velocity versus time for world records in running and swimming over various distances, a hyperbolic relationship between work rate and time exists. The point at which the power curve intensity/duration distribution begins to level out, mathematically termed the *asymptote*, can be termed critical power (figure 4.3). Thus, critical power can be described as one's sustainable exercise intensity, roughly over 30 to 60 min. Physiologically speaking, this tends to correspond to an individual's maximal lactate steady state, or an exercise intensity at which blood lactate production appears balanced by its removal (see chapter 3).

Anaerobic Work Capacity (W')

Anaerobic work capacity (W'), sometimes referred to as anaerobic speed or power reserve (ASR/APR), describes a theoretically finite or limited energy supply available above critical power. Figure 4.3 attempts to illustrate the W' /ASR/APR concept using a number of different-sized boxes across the various duration and intensity combinations above the critical power. Some boxes are tall and skinny, some are square and fat, and others are long and thin. What is interesting, however, is that the size or volume of each box is the same. The box size or box volume is our theoretical W' and represents a finite source of stored energy, theoretically speaking, the stored anaerobic alactic (ATP/CP) and anaerobic glycolytic energy (chapter 3). What this tells us is that when we're performing exercise above our critical power, as we do with HIIT, we only have a finite amount of that energy to play with before we need to recover. While there are a number of different ways we might use that energy (i.e., boxes 1 to 4 in figure 4.3), we only have a limited amount.

We can equate W' to a battery, which has a limited amount of stored energy (figure 4.4). When we exercise above our CP, we have to use some of this energy. Our battery runs down. However, if we allow a period of recovery, we can restore some of our W' battery.

Recovery Period

The recovery period between intervals receives much discussion from sport commentators and coaches alike. The most common belief we have seen purported is that recovery from HIIT should be active to promote the removal of the noxious blood lactate that has accumulated in the muscle during the high-intensity work. But is that really the case?

The benefit of active recovery has often been assessed via changes in blood lactate concentration (3, 6), which has little to do with muscle lactate concentration (40). Additionally, neither blood (34, 58) nor muscle (40) lactate has any direct (or linear) relationship with performance capacity. Thus, there is little support from a scientific standpoint for the sport commentator's typical recommendation to keep recovery active so as to clear the noxious blood lactate. While long-held myths rarely change in a lifetime, this is an important concept to grasp as a student, coach, or practitioner going about your craft of solving the HIIT-performance puzzle.

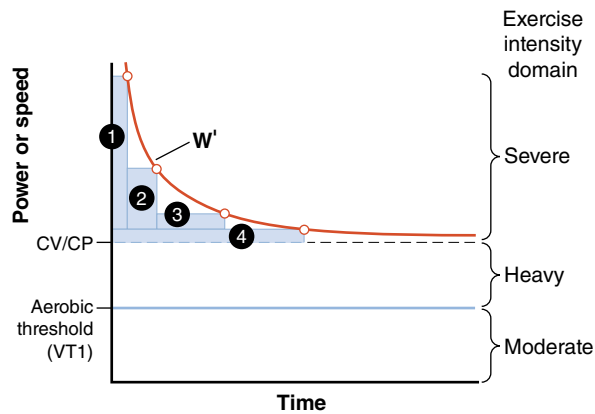


Figure 4.3 Critical power/velocity (CP/CV) and anaerobic work capacity (W'). The concept of W' is described further using the boxes numbered 1 through 4, which all show different usages of W' across varying intensity/duration combinations. Importantly, despite the different intensity/duration combinations, all W' boxes display the same total finite volume. VT_1 : ventilatory threshold (aerobic threshold; see chapter 3).

Adapted by permission from A.M. Jones, A. Vanhatalo, M. Burnley, R.H. Morton, and D.C. Poole, "Critical Power: Implications for Determination of $\dot{V}O_{2\max}$ and Exercise Tolerance," *Medicine & Science in Sports & Exercise* 42 no. 10 (2010): 1876-1890.

A better way to look at the recovery interval intensity and duration dynamics may be to consider it from the perspectives outlined in the HIIT work interval and duration description (figure 4.3). That is, we have a relatively finite energy source and need both time and certain conditions to optimize its repletion. A number of physiological aspects, such as those outlined in chapter 3, will influence this repletion of potential energy in muscle during the recovery period. One of the key factors is the quantity of a muscle's phosphocreatine (PCr) stores, sometimes called the short-term energy system. PCr is available in muscle to rapidly restore ATP levels. When PCr levels are high, so too is W' , and vice versa.

Thus, a period of recovery allows our system to recharge our W' battery to perform more HIIT, which may provide important adaptation signals. The recovery dynamics are determined by the chosen recovery period intensity and duration.

The recovery period intensity and duration share a similar energetic relationship with the exercise period intensity and duration. That is, we'll recover quicker in a given time period when the intensity is lower, and additionally, we can recover further when the recovery duration is prolonged. Looked at from the other angle, we can increase the overall workload and metabolic rate of a given HIIT session when we

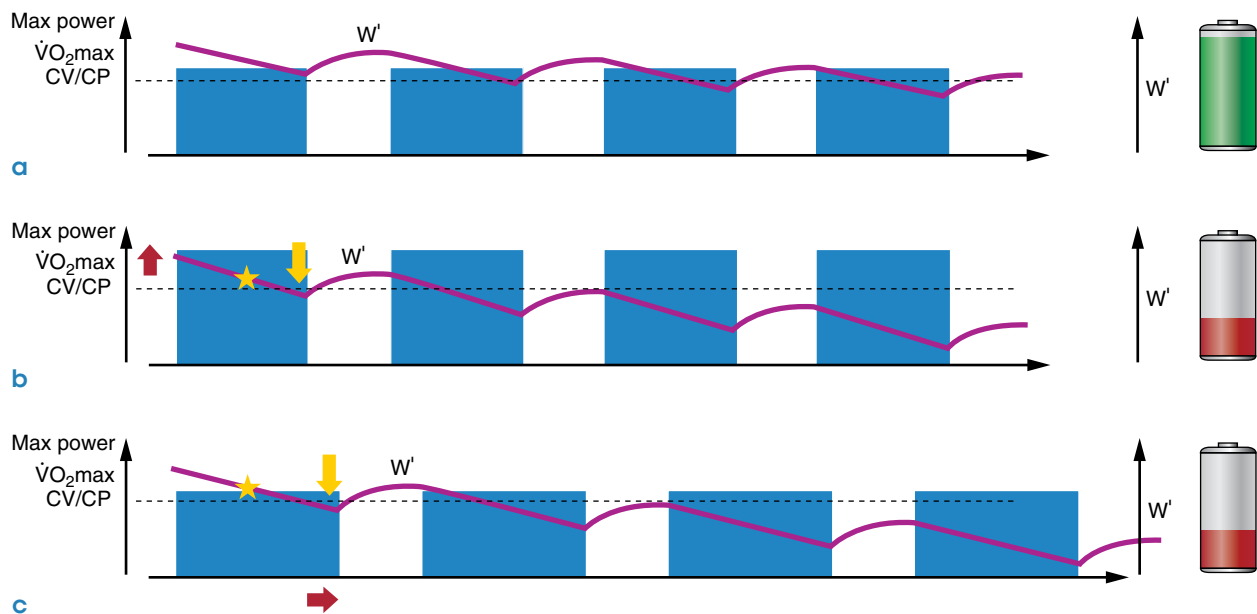


Figure 4.4 Finite W' being depleted with HIIT. The recovery period facilitates recovery of the W' . (a) An appropriate HIIT session for an individual in which W' is managed. (b) An HIIT session in which repeated bout intensity depletes W' toward a minimum (fatigue). (c) An HIIT session in which repeated bout duration depletes W' toward a minimum (fatigue).

either increase the recovery intensity (toward critical power) or lessen our recovery duration.

To illustrate the point, Dupont et al. (23) compared the effects of active (40% $\dot{V}O_2\text{max}$) versus passive recovery on how long subjects could repeat a 15 s/15 s intermittent high-intensity exercise sequence. Time to exhaustion for intermittent exercise with passive recovery (962 ± 314 s) was more than two times longer compared to the active recovery condition (427 ± 118 s). Thus, passive versus active recovery made a massive difference to the energy these subjects had available to perform their short-interval HIIT sessions. W' was more protected. Looking at the flip side of the coin, average metabolic power during intermittent exercise with passive recovery was marginally lower compared to the active condition (48.9 ± 4.9 vs 52.6 ± 4.6 mL·kg⁻¹·min⁻¹). To explain these results, the authors measured oxyhemoglobin saturation (SaO_2) via near-infrared spectroscopy (NIRS) and showed that the mean rate of SaO_2 decrease was lower with passive recovery versus active recovery (figure 4.5). Thus, more available oxygen ultimately means better recovery of W' .

We also examined this concept using a repeated sprint training exercise format that is also in line with what team sport athletes in the field use, and compared the effect of active versus passive recovery on all-out running performance and physiological markers in male team sport athletes. Subjects performed six repeated maximal 4 s sprints interspersed with 21 s of either active (2 m/s) or passive (standing) recovery on a nonmotorized treadmill. Running speed was lower and speed decrement was greater when recovery was active versus passive. Additionally, oxygen uptake, blood lactate, and deoxyhemoglobin were higher, indicating a greater metabolic demand for active versus passive recovery with HIIT (15) (figure 4.6). Again, passive recovery clearly defended W' better than active, allowing greater performance in subsequent bouts.

These studies highlight some opportunities where knowledge of the recovery intensity and duration kinetics can allow us to skin our cat optimally. For example, if we are after maximal recruitment (and likely adaptation signal) to our larger motor units, where engagement of such fibers is important, recovery should be maximal (i.e., exclusively passive over short durations) to allow for a higher reoxygenation of myoglobin, a higher PCr resynthesis, and a greater return of our W' . If practitioners still prefer the active recovery component (keeping athletes active or busy

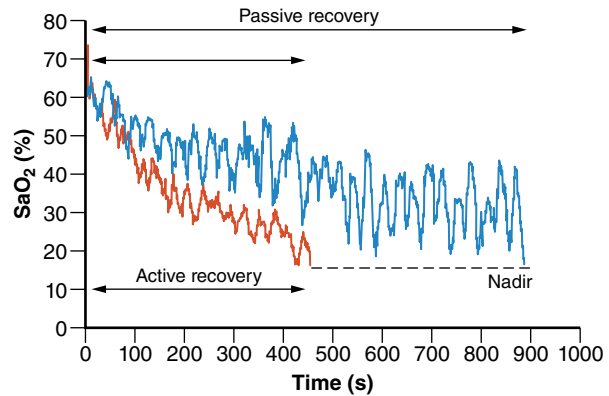


Figure 4.5 Effect of active versus passive recovery during a 15 s/15 s HIIT sequence to exhaustion. Passive recovery allowed subjects to double their exhaustion time, due in part to higher oxyhemoglobin saturation (SaO_2) and lower metabolic rates compared with the active recovery (40% $\dot{V}O_2\text{max}$) condition.

Reprinted by permission from G. Dupont, W. Moalla, C. Guinhouya, S. Ahmaidi, and S. Berthoin, "Passive Versus Active Recovery During High-Intensity Intermittent Exercises," *Medicine & Science in Sports & Exercise* 36 no. 2 (2004): 302-308.

alongside aerobic involvement), the detrimental effect of recovery exercise intensity on W' can be compensated for by using a longer duration. Additionally, if we are time poor, or if a practitioner has a limited window to apply a metabolic stimulus, and total metabolic rate in a given time period is important, we can also use active recovery to raise the total metabolic work rate, depleting W' faster and raising $\dot{V}O_2$ and lactate.

Figure 4.7 provides a theoretical framework for understanding the energetics of the recovery period. When the recovery intensity is low, the rate of W' recovery is fast, and vice versa. Likewise, when the recovery duration is low, the repletion of our depleted W' is limited.

As shown theoretically (figure 4.7), both the duration and intensity of the relief interval are important, and directly impact on the repletion of W' and subsequent physiological effects of the HIIT session (45). Both the duration and intensity of the relief interval must be considered in light of:

1. Maximizing work capacity during subsequent intervals. Since active recovery can lower muscle oxygenation (15, 22), impair PCr resynthesis (O_2 competition), and trigger anaerobic system engagement during the following effort (50), it may be recommended to

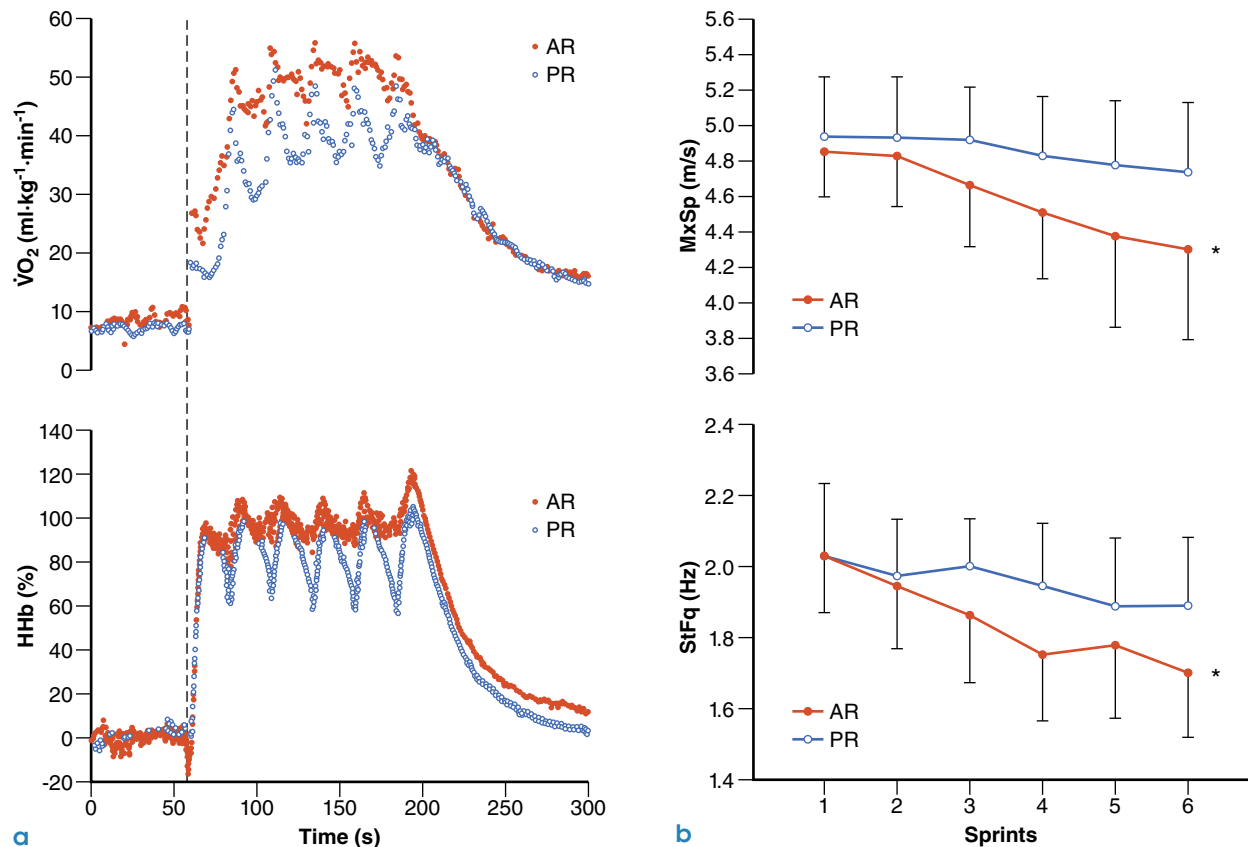


Figure 4.6 (a) Mean $\dot{V}O_2$ and deoxyhemoglobin (HHb) expressed as a percentage of HHb level and (b) maximal speed (MxSp) and stride frequency (StFq) during six all-out sprints interspersed with 21 s of either active (AR) or passive recovery (PR). Values are means $\pm$ SD ($n = 10$).

Data from M. Buchheit, C. Cormie, C.R. Abbiss, S. Ahmaidi, K.K. Nosaka, and P.B. Laursen, "Muscle Deoxygenation During Repeated Sprint Running: Effect of Active vs. Passive Recovery," *International Journal of Sports Medicine* 30 no. 6 (2009): 418-425.

use passive recovery typically to allow the maintenance of work quality, and in turn, a longer T_{lim} (i.e., figure 4.7a).

2. Maintaining a minimal level of $\dot{V}O_2$ to reduce T to $\dot{V}O_{2max}$ during subsequent intervals (i.e., starting from an elevated baseline) (7, 44). While performing active recovery between interval bouts is appealing to accelerate T to $\dot{V}O_{2max}$ and, in turn, induce a higher fractional contribution of aerobic metabolism to total energy turnover (21), its effects on performance capacity (T_{lim} , and hence, T at $\dot{V}O_{2max}$) are not straightforward. In fact, during HIIT with short intervals, active recovery impairs W' recovery and, in turn, shortens T_{lim} and, in turn, T at $\dot{V}O_{2max}$ (figure 4.7c). While a beneficial performance effect on subsequent intervals can be expected

with long recovery periods (≥ 3 min) (8, 19, 21) when the possible washout effects overcome that of the likely reduced PCr resynthesis, active recovery performed during this period may also negate subsequent interval performance using both long periods at high intensities ($>45\%$ v/p $\dot{V}O_{2max}$) (6) and short periods of varying intensity (22, 51). Again both may compromise T at $\dot{V}O_{2max}$ (figure 4.7, b and c). In the context of long-interval HIIT, passive recovery is therefore recommended when the relief interval is less than 2 or 3 min. If an active recovery is chosen for the above-mentioned reasons (7, 21, 44), relief intervals should last at least 3 or 4 min at a submaximal intensity (6) to allow the maintenance of high exercise intensity during the following interval.

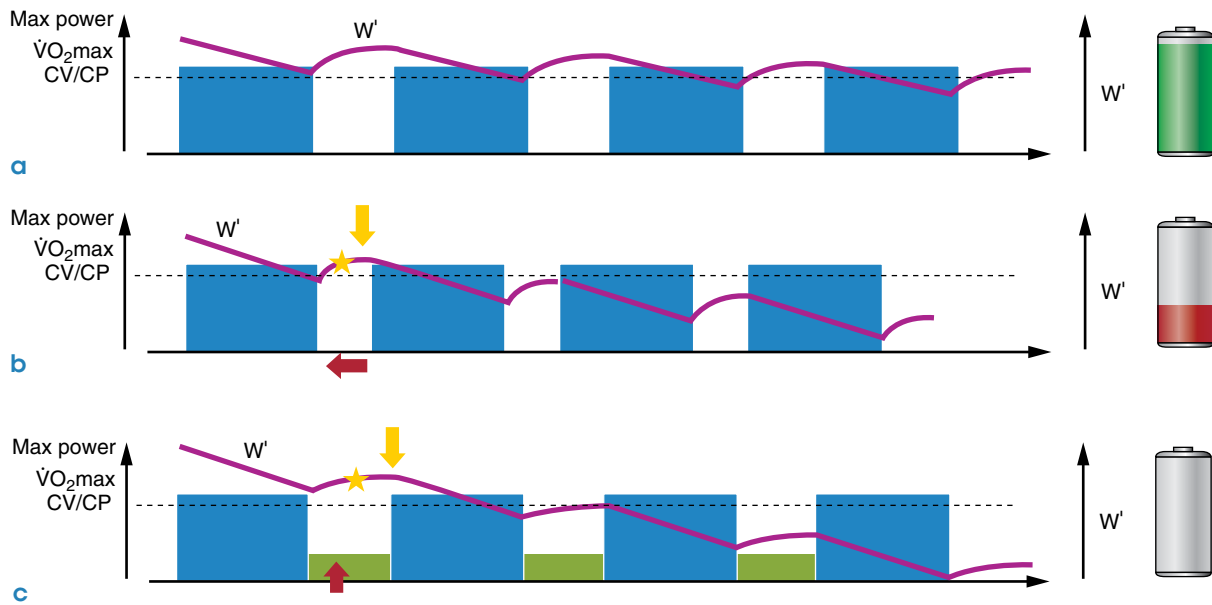


Figure 4.7 Influence of shortened duration and active recovery on W' repletion during HIIT. The recovery period facilitates recovery of W' . Blue represents the HIIT. Note that intensity is greater than the critical velocity/power (CV/CP). Purple line is W' . (a) In this HIIT session, W' is managed well. (b) The recovery duration is shortened and the time is not sufficient enough to recover W' substantially. (c) An active recovery condition, in which the heightened metabolic rate reduces even further the speed of W' recovery.

3. Maintenance of a high cardiovascular strain. As described in chapter 3 (figures 3.4 and 3.5), within the context of long-interval HIIT, stroke volume remains high with passive recovery, which may facilitate the signal to enhance cardiac output during HIIT (52). Recall too that the HR response during the recovery phase of HIIT is neither related to systemic O_2 demand nor muscular energy turnover (14, 59) but rather to central nervous system and metaboreflex responses (48). Thus, within any single HIIT session, the HR recovery response is somewhat irrelevant within the context of W' .

In summary, we have shown that the intensity and duration of the recovery phase has a substantial effect on both subsequent interval bouts and the HIIT session as a whole. If performance in repeated high-intensity bouts is important, recovery should be prolonged and passive (although recovery durations greater than 4 or 5 min make active recovery a good option also). If overall workload in a given time period is a priority, as it can be when a practitioner has only a short time window to work with, recovery intensity can be heightened or recovery

duration shortened to enhance overall work completed in a given time period and the associated physiological response.

In practice, active recovery is psychologically difficult to apply for the majority of athletes, especially for nonendurance athletes. When moderately trained runners ($v\dot{V}O_{2max} = 17.6$ km/h) were asked to self-select the nature of their relief intervals during an HIIT session (6×4 min running at 85% $v\dot{V}O_{2max}$ on a treadmill with 5% incline), they chose a *walking* recovery mode of about 2 min (49). Compared with 1 min recovery intervals, the 2 min recovery duration enabled runners to maintain higher running speeds; extending passive recovery to 4 min did not provide further benefits with respect to running speeds. Thus, best practice for the majority of long-interval HIIT appears to be 2 min of passive rest or easy walking.

Number of Intervals or Series Duration

When we refer to series duration, we're ultimately referring to the number of interval repetitions or HIIT bouts we're performing in a given training session (figure 4.2). In either case, the influence of these

parameters relates to the total metabolic draw on the system. The more an HIIT session goes on, the larger the aerobic system draw is required, as the short-term systems (CP/glycolytic/ W') are taxed more and more, without adequate recovery time (unless recovery time is passive and prolonged; figure 4.7). Thus, the series duration is another way that we can skin the cat to affect the response of the training session. If we shorten the series duration (i.e., number of bouts), we generally enhance the metabolic rate or quality within a given time period, which might be appropriate in the context of allowing us to focus on other aspects of performance, such as for team sports where technical and tactical training may be prioritized. Alternatively, we can lengthen our series duration, drawing on aspects of endurance or fatigue resistance as per the needs of our more endurance-based athletes (figure 4.8).

Within the team sport context, we might use a number of shorter series, so that for a similar total number of repetitions, we achieve a higher volume of quality work. As illustrated in figure 4.8, for example, two periods of 4 min bouts separated by a longer recovery period (figure 4.8a) might be better

than a single 8 min period of HIIT bouts in the context of achieving quality or intensity (figure 4.8c), as in this latter example, fatigue ensues, which can impact on the neuromuscular response (chapters 3 and 6). Finally, there is a lower limit we should place on our series duration in the context of targeting aerobic adaptations. Series duration should be no less than 3 or 4 min to enable athletes to reach a $\dot{V}O_2$ plateau (or near $\dot{V}O_{2\max}$; see chapter 5). Generally, series durations are between 4 min (quality emphasis) and 14 min (endurance emphasis) (figure 4.8).

Number of Interval Bout Series (Total Volume)

As with our series duration (number of intervals; figure 4.2), the series number and total volume have a similar effect. The greater the number of series, the longer the overall session will be, and similarly, the greater the workload this creates (see again figure 4.8 a, b). The effect here again generally is to push the aerobic adaptation stimulus, unless recov-

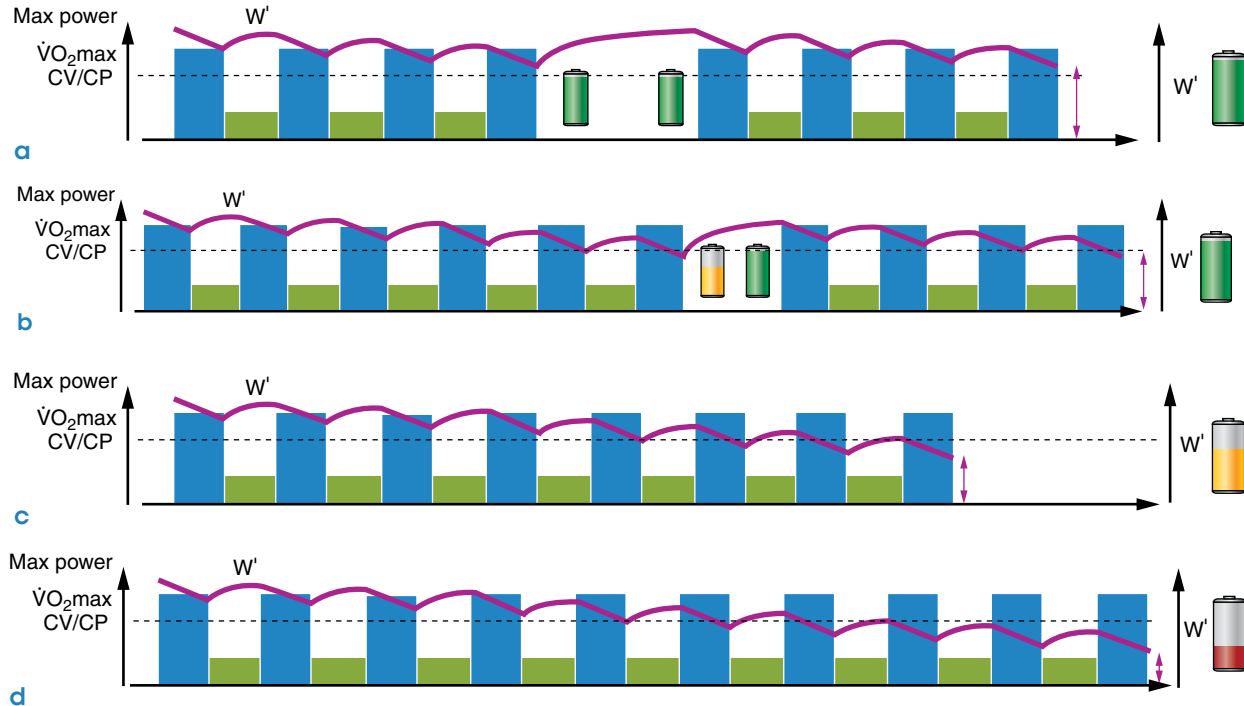


Figure 4.8 Theoretical impact of varying the series duration. (a) W' is maintained in this 2 × (4 × 4 min bouts) HIIT session separated by a single series recovery period. (b) Progression to a slightly more taxing session. (c) Alternate to session A, without the series recovery period (i.e., 8 × 4 min). (d) An excessively long series duration without adequate recovery.

ery duration is long and passive. Conversely, reducing the series number lowers the total volume of the training load.

This variable is essentially related to our training load, be that kilometers traveled or power produced for a given time, since the quantitative metabolic response can be equated between varying series durations depending on the recovery type (for elaboration, see chapter 8).

Between-Series Recovery Intensity and Duration

As per our recovery period intensity and duration (figure 4.2), the same rules apply. If we lower our between-series intensity, we'll speed the recovery of W' . Likewise again, if we lengthen the between-series recovery duration, we'll amplify W' recovery. Conversely, a raised between-series recovery intensity or shorter duration will lower the W' recovery rate and raise the metabolic rate within a given period of time (figure 4.9). Consequently, just as with our recovery period intensity and duration, we can facilitate either more quality work in a subsequent training series (with more passive and longer recovery) or achieve a higher metabolic rate in a given time period in the case where we are time starved and attempting to achieve a higher aerobic workload for a given period.

Total Work Performed (Volume)

To this point, variables 1 through 8 make up the ninth parameter, the total work performed in an HIIT session, or the session volume. This total amount of work or volume, a product of the session's overall intensity and duration, can be quantified in a number of ways (chapter 8) to enable appropriate progression of the training program parameters throughout a given training cycle. Knowledge of the total work performed can assist to prevent common pitfalls in which athletes perform too much HIIT work without appropriate recovery, which can lead to overtraining (chapter 7).

Exercise Modality

In general, when not specified, HIIT exercise modality tends to be that involved with the sport of interest, e.g., running for run-based sports, cycling for cyclists, boxing for boxers, rowing for rowers, etc. There are, however, many occasions when, for example, typical HIIT exercise modalities can be modified to adjust the acute metabolic or neuromuscular responses. In fact, when we refer to exercise modality in the context of an HIIT session (figure 4.2), we directly refer to the different ways we can manipulate the session to adjust the locomotor, neuromuscular, and musculoskeletal strain on the body. We originally

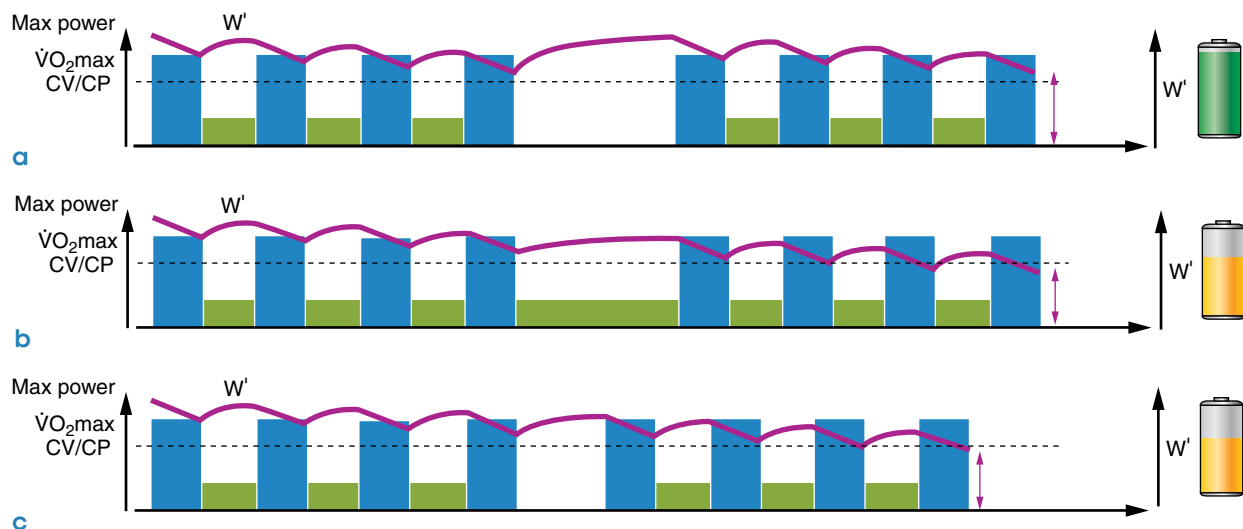


Figure 4.9 Between-series recovery intensity and duration effect. (a) Between-series recovery intensity is passive and prolonged, which enhances recovery and returns W' adequately. (b) Between-series recovery intensity is higher (active), resulting in less recovery and associated W' restoration. (c) Between-series recovery duration is shortened, which lowers between-series recovery and W' .

discussed this in chapter 3, and note that this can be implemented irrespective of the primary sport that the athlete participates in. On one end of the spectrum, we have the gross differences in mode, comparing across exercises like swimming, running, rowing, and cycling. However, other more subtle changes in exercise mode can be made to alter the training stress and stimulus: grass versus sand running; shuttle versus straight-line running; uphill versus downhill running; as well as cadence (cycling), bungy (rowing/kayak), and band, paddle, or sponge work (swimming).

Manipulations of the exercise mode can be used to target specific areas or regions of the body that need more or less locomotor, neuromuscular, or musculoskeletal strain versus (or in addition to) metabolic strain. In specific occasions for substitute soccer players, such as when increased neuromuscular load is required to compensate for the lack of on-field playing minutes, we may have them perform run-based sessions with more CODs than they'd usually encounter. Conversely, starting soccer players might perform treadmill HIIT sessions for the maintenance of central cardiovascular capacity alongside minimizing musculoskeletal strain that might be attained in matches. Heavy rugby players might perform low eccentric cycling HIIT sessions or even boxing-based HIIT to off-load their legs in the 2 to 4 d following a match, while cyclists might perform a swimming-based HIIT session as a means of achieving more whole-body aerobic exercise with

reduced lower leg involvement. For an 800 m runner showing evidence of low resiliency to musculoskeletal strain, a block of hill runs might be prescribed in the preparation phase to enhance this aspect, irrespective of the metabolic targets of the session (figure 4.10). Mode-specific effects (i.e., uphill versus straight-line running) on aerobic adaptation targets are expanded on throughout chapter 5, and a decision process flowchart to assist the practitioner with programming decisions is shown in figure 4.10.

Finally, it is also worth noting that many practitioners in team and racket sports use more sport-specific exercise modalities as alternatives to run-only efforts, e.g., tennis players hitting balls (chapter 18) or soccer players passing and shooting during short intervals (chapter 30). While the rationale for doing this still requires evidence-based support, the practice is believed to improve the magnitude of the HIIT-related physiological and locomotor gain transfer into the activity. Another benefit that is easy for everyone to appreciate is the greater motivation that athletes generally show when performing such HIIT sequences in a sport-specific fashion. In these particular drills, however, athletes are generally repeating closed skills with little decision-making and interactions with opponents, which questions its actual specificity and differentiates these HIIT exercises from proper game-based drills that involve aspects of playing, gaming, or puzzle solving (see chapter 2).

It is worth mentioning the widely used resistance-based HIIT circuit work that involves several different

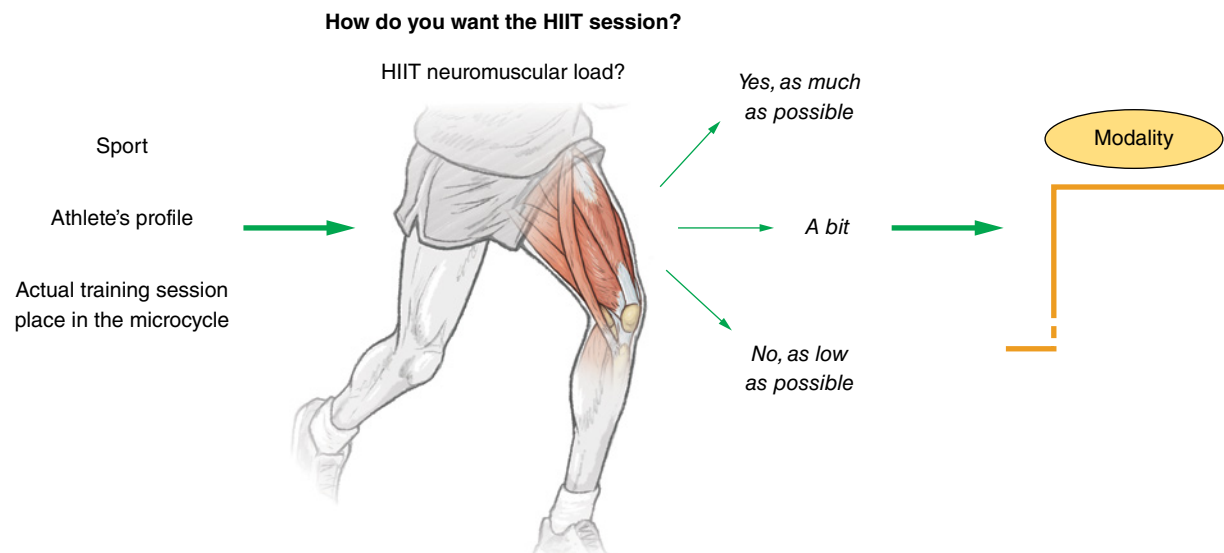


Figure 4.10 Ways to change modality, which likely affects musculoskeletal strain and neuromuscular load.

strength exercises during each exercise interval (e.g., generic exercises such as push-ups, squats, dips, crunches, lunges, etc.), repeated for 15 to 30 s, interspersed with passive rest (often with a 1:1 ratio), which might be considered as some of the most extreme forms of neuromuscular-oriented and demanding HIIT sequences known (type 4). Far from being specific to any sport, such exercise routines are believed to be a key component of a general physical preparation program, targeting both the cardiovascular and neuromuscular systems, especially during the initial stage of an athlete's program (e.g., pre-season, building-up phases). Historically, these exercises were performed at a relatively low intensity (40% to 60% 1RM); however, the more recent explosion of circuit-training-related programs has extended to the use of heavy loading (>60% to 80% 1RM), which requires lower work:rest ratios. While research has shown that such HIIT sequences can be used to concurrently improve aerobic function (i.e., $\dot{V}O_2\text{max}$) and muscle strength (i.e., 1RM), benefits of such a training approach are likely to be exclusive only to individuals with limited initial fitness levels (i.e., the lower the initial fitness level, the greater the chance for $\dot{V}O_2\text{max}$ and 1RM improvements). Our contrasting view today (46), in alignment with our philosophy set out in chapter 1 (figure 1.5), is that trained athletes need to be trained like Special Operation Navy Seals (13). Indeed, neither the metabolic nor the strength/power stimulus of such sequences would be considered intense enough to promote the desired adaptations for the elite athlete. We believe the toothpaste theory nicely illustrates the concept. The circuit-based approach can be likened to squeezing toothpaste only from the middle of the tube and is useful strategy for only a given period. Highly trained athletes require a more-focused target approach (i.e., separate HIIT and power, strength, and speed work) to get all the toothpaste out. While resistance-based HIIT circuits do a good job of squeezing from the middle of the tube, after some time they are an ineffective strategy for achieving any sort of outcome to get the toothpaste out, while squeezing strongly and consistently on just one side, and before the other, allows a much more efficient means of getting toothpaste on our brush.

Finally, we also feel that the practice of performing strength exercises during the work interval of an HIIT session may be at odds with the generic principles of strength exercise prescription. While the participation of the different energy system response targeted for metabolic conditioning is exer-

cise duration dependent (which justifies the use of predefined work interval durations, see exercise period duration), strength exercises should be prescribed by a given number of repetitions in relation to the load and not by series duration (9).

Environment

The environment where HIIT is performed, namely the ambient temperature and humidity or atmospheric pressure, can create profoundly different physiological effects (30). Clearly we can push the envelope in this area by creating greater physiological stress in athletes over what they would typically have in normal training to enable them to be more fatigue resistant. Changes in environmental conditions can also be used to increase the internal:external load ratio, so that low mechanical work can be associated with high metabolic cost, which can be useful to limit neuromuscular load during an athlete's return to full training after an injury or a team's busy schedule. Of course, this is only if the training program is well managed. The two main environmental moderators include the environmental heat load (temperature and humidity) and the partial pressure of oxygen (altitude).

Heat Load

Heat load in the body can be altered in a number of ways. Heat load is a sum of heat gain and loss factors. Heat is gained via the body's own metabolic heat production (the breakdown of foodstuffs is relatively inefficient, and roughly 80% of the energetic process used to make energy is lost in the form of heat) along with the environmental heat load if greater than skin temperature. Heat is lost through the processes of convection, conduction, radiation, and sweat evaporation. In any condition where heat is gained, it becomes stored, and is stored largely in the core of the body (termed *core temperature*).

The rise in core temperature has a few effects that we'll all be very familiar with. First, we know it is uncomfortable, and the natural response is to lower exercise intensity in an attempt to control metabolic rate and lower the heat load (2). The heightened body temperature redistributes total body water to the skin for cooling via sweat evaporation, and heart rate is increased to maintain cardiac output and blood supply to working muscles (20).

While initially posing challenges to performance, the situation can create a number of positive effects that can be beneficial (30). First, the acutely lowered

plasma volume creates a rebound effect and subsequently raises plasma volume in the recovery period (53), which can improve cardiovascular control and increase stroke volume. Additionally, the higher heat load creates increased energy demands that are initially met through increases in glycolysis during muscular contraction (26). This raised AMPk signaling increases the activity and amount of peroxisome proliferator-activated receptor gamma coactivator 1 alpha, a protein described in chapter 3 that can directly induce the primary adaptive responses to endurance exercise: mitochondrial biogenesis, angiogenesis, and increases in fat oxidation (5). Heat shock proteins (HSPs) are a group of proteins, the expression of which is increased when the cells are exposed to elevated temperatures. The increased activity of HSPs makes these cells more resilient to subsequent insult or heat exposure (36). Thus, environmental heat load can be manipulated a number of different ways that provide various potentially positive physiological effects (27).

Raised heat load also makes training feel harder and comes with added sympathetic nervous system stress. Rating of perceived exertion is increased alongside thermal discomfort. This all occurs without the need to increase the mechanical (neuromuscular and musculoskeletal) load in the body; in fact it can often be lower. Indeed, we typically find a downregulation of neural drive in such conditions, as HIIT becomes more difficult (1). This could be particularly the case with very hard efforts, such as sprint interval training, where achieving an all-out effort might be important (32, 33). Alternatively, warmer muscles from hotter temperatures can in some cases enable heightened levels of neuromuscular recruitment and associated physiological responses (16).

While we've highlighted the main adaptive benefits and opportunities, that is, heightened cardiometabolic responses, attention to tactical and technical aspects of sport and learning may be compromised and lowered central drive can also be problematic for training at certain times. Thus, appropriate timing of heat load before, during, and after HIIT is important and needs to be managed appropriately around context. One example often used across a number of sports is to perform an HIIT session in the heat to increase the metabolic load under low neuromuscular impact (HIIT type 1 response). Chapter 5 will expand on how we can use this factor to skin the cat appropriately to refine our weapons and hit our targets.

Hypoxia and Altitude

Another environmental factor that we might manipulate to skin the cat is the level of hypoxia an athlete's body is under. There are many ways this might be achieved, reviewed extensively by Girard and colleagues (31), including various forms of altitude training (both natural and simulated means), including local occlusion methods, as well as voluntary hypoventilation or breath holding (56). It's important for the reader to appreciate here that we are specifically only referring to the acute cardiometabolic responses that might be occurring in a single HIIT session, and not the chronic systemic responses that may occur with long-term training strategies at altitude where the aim is directed more toward achieving enhanced hematopoiesis (more red blood cells).

The main acute response of interest for hypoxic HIIT is that of its effect on enhancing enzymes of glycolysis, mediated by the oxygen-sensitive hypoxia-inducible factor (HIF)-1 α (25) (figure 4.11). While the effects of hypoxic training on HIIT completed in hypoxic chambers has generated a number of impressive and insightful works of research from the laboratory of Professor Gregoire Millet and coworkers (10, 24), the practice of using artificial means of generating hypoxia is less common and typically reserved for athletes with access to specially designed facilities. Such altitude chambers (normobaric) are expensive and not always convenient to access for most athletes and teams. Moreover, accessing regions of true altitude (i.e., elevations between 1500 and 2200 m) is also not often convenient, and typically costly. Importantly, while there is evidence for a marginal benefit to training repeated sprint ability in hypoxic relative to normoxic conditions across studies (standard mean effect size=0.46, 95% CI -0.02 to 0.93) (10), and although being used by many individual athletes and even some teams, the strategy of using hypoxic facilities has, in our opinion, a highly questionable cost:benefit ratio. Use of vascular occlusion methods, while perhaps used in certain individual cases, is also difficult for us to recommend based on the minor effects relative to the time and effort involved (relative to other factors in most athletes' performance puzzles). The one exception that has a certain level of practicality in our opinion might be sets of voluntary hypoventilation training, such as those sometimes performed in swimming HIIT for swimming and triathlon.

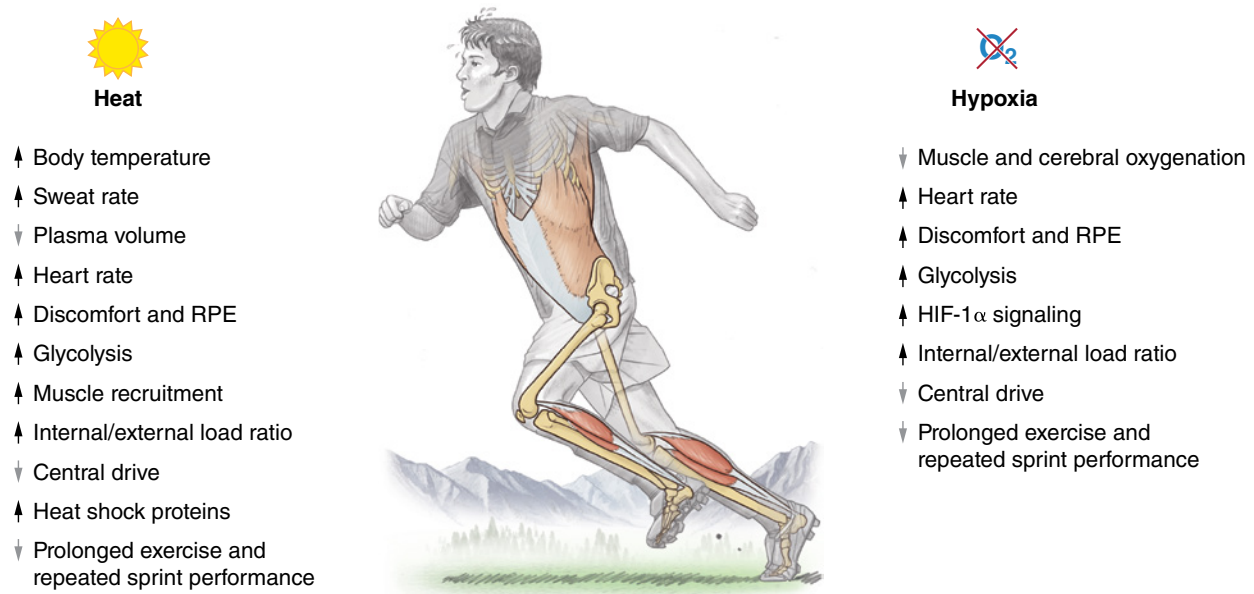


Figure 4.11 Acute impact of hot and hypoxic environments during HIIT.

Regardless, when athletes either reside in or travel to altitude and perform HIIT, there are similar physiological effects as those shown with heat training (30). For example, the lower partial pressure of oxygen in ambient air reduces the driving force of oxygen to bind to hemoglobin, lowering oxyhemoglobin saturation. The downstream effect is a lower partial pressure of oxygen in tissue, which targets the classic peripheral aerobic adaptations in muscle needed to provide enhanced energy needs, including increases in capillary density, myoglobin content, and oxidative enzymes. Additionally, heart rate is increased for the same absolute training intensity to facilitate the metabolic needs of tissues. Thus, we often observe reduced power and speed with prolonged efforts, and therefore enhanced acute metabolic responses relative to musculoskeletal and neuromuscular strain in normoxia. The effect on short-duration, high-intensity performance is unaffected, unless efforts are repeated (31). Thus, consideration for inter-bout and inter-series recovery intensity and duration is needed, depending on the training objectives. Additionally, as with heat stress, hypoxic stress can lower central drive, potentially diminishing training quality or influencing technical and tactical skill acquisition. On the one hand, this could potentially appear problematic due to the heightened cognitive challenge, while on the other hand, sometimes conditions made harder (through either altitude or heat) might facilitate heightened performance levels when the stress is removed. An

additional consideration for team sport players at altitude is the effect that the lower atmospheric pressure has on ball trajectory and associated task or skill performance. Thus, effects on skill or motor learning needed for team sport athletes need to be weighed against the possible physiological effects of training in such environments.

In summary, both heat and altitude can be used within the HIIT programming puzzle to affect the internal exercise cost to external power production ratio and the associated adaptive acute response to exercise training. Both heat and hypoxia are effective means of enhancing the acute metabolic response during exercise, which can be useful for enhancing the metabolic requirement of an exercise bout relative to the neuromuscular and musculoskeletal strain. As well, various means of creating hypoxic conditions during repeated sprinting appear effective at enhancing repeated sprint ability and glycolytic enzyme activity, although such training should be considered based on a cost:benefit ratio.

Nutrition

It's no secret that we all have to eat to survive. You wouldn't be reading this otherwise. Nutrition provides needed energy, the water that makes up the body, as well as everything we need to run our bodies (micronutrients, etc.). While this book is not about nutrition, nutrition clearly has an influence on the physiological effect of an HIIT bout, which is why

we are covering it. While there are likely other subtle effects that can be caused by manipulating other dietary nutrients, including supplements, we will focus on what we believe are the key nutrient manipulations influencing physiological effects. These include the dietary carbohydrate availability, hydration status, and caffeine intake. Chapter 7 additionally explores some of the common pitfalls and detrimental effects a poor diet has on performance. Like environment, nutrition is an additional area that can help us make further benefits in training, if well manipulated.

Carbohydrate Availability

Carbohydrates are plant-derived sugars that come in a variety of forms. At the simplest level, and the level to which all carbohydrates are broken down eventually, is the compound glucose, a six-carbon molecule. Glucose is a primary metabolic fuel used by all cells to give us energy. Specifically it is the fuel of use for the process of anaerobic and aerobic glycolysis (chapter 3).

Notwithstanding the profound individual variance in the blood glucose response to the consumption of foodstuffs (64), there are some general principles we might appreciate around the topic of nutrition, acknowledging that refinement and optimization will exist at a monitored individual level. In the context of most athletes habituated to a mixed-Western diet high in carbohydrates, the act of lowering carbohydrate availability can create stress on (among others) muscle and nerve cells. The resultant lowered level of blood glucose, termed *hypoglycemia*, acutely creates a perceived feeling of fatigue and a reduced motivation to exercise. Ultimately this global effect of the lowered basal blood glucose level reduces systemic central nervous system activation. Under such conditions, acute high-intensity exercise capacity is typically reduced (35).

While high-intensity exercise capacity is typically lowered in the acute state, such a state can create an advantaged adaptive condition. Lower blood glucose concentrations reduce glycolytic capacity, which drives AMPk signaling for associated benefits (mitochondrial biogenesis and enhanced fat oxidation rates; chapter 3). The condition also makes training harder, without increasing mechanical (musculoskeletal and neuromuscular) load, which is something we often want during a training session. Alternatively, the condition can be nonoptimal in the context of the training condition, and could

acutely impair aspects such as mood, confidence, coordination, agility, judgment, game intelligence, learning, and so forth.

A number of methods can be used to lower carbohydrate availability. The most obvious is to lower carbohydrate consumption in the diet. This can be challenging for many athletes as it requires nutrition knowledge, and as mentioned, the individual response to foodstuffs is extremely variable. The social aspect of food (i.e., habits, eating what others eat, eating what someone serves you, etc.) and access to appropriate foods (low-carbohydrate foods can be difficult to source in many places) additionally make this intervention difficult. Typically, when lowering carbohydrate content from the diet, additional calories need to come from somewhere, and generally speaking this largely will come from fat, a concept that can be difficult for many to appreciate based on tradition, ingrained mind models, or instilled beliefs surrounding the effect of dietary fat on health. Regardless, a number of methods can be used to manipulate systemic levels of carbohydrate content or blood glucose, including eating low-carb high-fat (LCHF), fasting, and sleeping low.

Low-Carbohydrate High-Fat (LCHF) Diet

The low-carbohydrate high-fat diet is just what it sounds like. With protein levels similar to that of a typical mixed Western-based or high-carbohydrate diet (about 20%), fat calories are increased toward 70%, with carbohydrate content falling to 10%, or below 100 g in a given day (57). The LCHF diet has received considerable attention in social media, but few studies have assessed its effects. In most studies that have examined the impact of an LCHF diet on high-intensity performance, the time course has been relatively short (3 d to 3 wk). For example, when Havemann et al. (35) examined high- versus low-carbohydrate diets over 6 d in cyclists, overall 100 km performance time was not different; however, high-intensity 1 km sprint performance during the time trial was impaired on the high-fat relative to the high-carbohydrate diet. As well, the authors found significant increases in HR and effort perception with the LCHF diet after 6 d. However, as with training adaptation, which requires long periods of adaptation, dietary adaptation may follow a similar time course. To address this discrepancy in the literature, Cipryan et al. (18) recently had recreationally trained participants change from their habitual mixed Western-based

diet to a ketogenic (<50 g CHO) diet over a 4 wk period. After this 4 wk period, performance and cardiorespiratory responses during a graded exercise test and HIIT were not impaired in the LCHF diet group. The LCHF group did, however, substantially increase their rates of fat oxidation. While more research is needed, it appears that short-duration time course adaptation periods can compromise HIIT performance, with adaptation requiring at least 4 wk. Additionally, LCHF diets substantially enhance fat oxidation, which is of importance for fatigue resistance and prolonged exercise performance (37).

Fasted Training

Fasted training is simply that—training in the fasted state. This can occur any time we decide not to eat for a given time period. Under the fasted condition, blood glucose and insulin levels drop and fat oxidation rates increase to meet the energetic demands. When we train under such a condition, we drive further the fat adaptations previously mentioned with the LCHF diet and with high-intensity training even more so (increased AMPk signaling (63)). Anecdotally speaking, this is a common approach to training for many athletes, especially endurance specialists (54).

Sleeping Low

When we feed in the evening (typical) and we go to sleep, we are in effect fasting. When we wake, still in the fasted state before breakfast, we can train without breakfast and we are in effect typically already 8 h fasted (54). Lowering carbohydrate content in the evening meal, which can subsequently lower muscle glycogen levels, termed *sleeping low*, can enhance the adaptations further. Indeed, Marquet et al. (43) have shown significant improvements in submaximal cycling economy, supramaximal cycling capacity, and 10 km running time in trained endurance athletes using such an approach.

Two-a-Day Training

We all know that athletes train, and many more than once in a day. When training occurs without refueling carbohydrate levels after the first session, muscle and liver glycogen levels may be only partially restored, and as a result, the same adaptations are driven in the subsequent session. As such, the signals for increased mitochondrial biogenesis (i.e., AMPk; chapter 3) and fat oxidation rates are enhanced (60, 62).

Hydration Status

When athletes train, specifically when they train in the heat, it's often emphasized that they need to maintain hydration status. While that might be true in some contexts, there are times in the healthy athlete context where exactly the opposite approach can enhance a physiological state. Indeed, a lowered hydration status, which acutely lowers plasma volume, creates stress in a number of different areas, including the heart and the kidneys (20). The endocrine system responds by retaining sodium at the level of the kidney (heightened aldosterone and arginine vasopressin levels), water follows, and plasma volume is increased. The enhanced plasma volume can be highly beneficial in a number of contexts, from creating partial heat acclimation to enhancing stroke volume through increased ejection fraction, thereby enabling cardiac stability. For example, Garrett et al. (29) showed that dehydration increased the desired adaptations from a short-term (5 d) heat acclimation protocol (90 min cycling in 35°C, 60% RH) in well-trained males. Aldosterone was enhanced more with dehydration, which was positively related to the plasma volume expansion, which tended to be larger in the dehydration condition. Thus, plasma volume expansion may be more pronounced with permissive dehydration and is one further factor that can be considered around HIIT to skin the cat. Of course, caution must be warranted when applying such maneuvers as physiological strain is likely to additionally be high, which could jeopardize other aspects of performance, such as skill development and tactical learning within the team sport context.

Caffeine

While there are, in our opinion, only a very small number of silver bullet solutions to solving the performance puzzle relative to the number of commercial claims, caffeine supplementation does show relatively consistent positive effects on exercise performance across a number of different conditions. In particular, caffeine couples well with train-low strategies to enhance acute HIIT performance. As described, under conditions of low-carbohydrate availability in athletes adapted to traditional Western diets (typical) or in any condition in which muscle glycogen content is substantially lowered (i.e., two-a-day training), HIIT performance (but not necessarily the acute signal/

response) is compromised (61). In any such condition (figure 4.11) when HIIT is planned, caffeine supplementation, typically taken in a variety of forms (including coffee and tea) approximately 1 h prior to the session, will enhance HIIT performance and potentially the acute metabolic signal in the larger motor units recruited. For example, Lane et al. (41) showed that caffeine ingestion ($3 \text{ mg} \cdot \text{kg} \cdot \text{BM}^{-1}$) taken 1 h prior to an HIIT session ($8 \times 5 \text{ min}$ maximal sustainable pace, 1 min recovery) conducted in a low-carbohydrate state enhanced power output by 3.5%. Thus, such a strategy may be particularly useful in athletes reluctant to train low for various reasons.

Figure 4.12 shows the main environmental and nutritional manipulations used to manipulate HIIT programming to assist practitioners to solve their programming puzzles.

In summary, we have outlined 12 different major ways that we can skin the cat around HIIT. With all those moving parts, how do we now take these and begin to make an impact on performance? How do we shape the puzzle piece so that it fits appropriately in the athlete's performance puzzle? How do we use this information to make useful weapons that will hit our physiological targets?

Skinning the Cat to Make Weapons

First, like soldiers going into battle, we need to make weapons if we are to take out our targets. Through trial and error and the experiences of many coaches, athletes, practitioners, scientists, and others who have gone before us, we feel there are five key weapons or formats we can form to induce the appropriate acute physiological response we are after to build our ultimate athletes. These formats include short intervals, long intervals, repeated-sprint training, sprint interval training, and game-based HIIT. These weapons can be used for different combat operations, depending on our physiological targets (figure 1.5).

Long Intervals

Long intervals are just that—long. In long intervals, we're using repeated bouts of the longer end of the intensity-time continuum around our $\dot{V}\text{pVO}_2\text{max}$ (95% to 105%) or 80% to 90% of $\dot{V}_{\text{IFT}}$. The durations need to be longer than a minute to induce the acute metabolic and neuromuscular responses we are after, which we will allude to in the next chapter. For long intervals to be effective, they should be separated

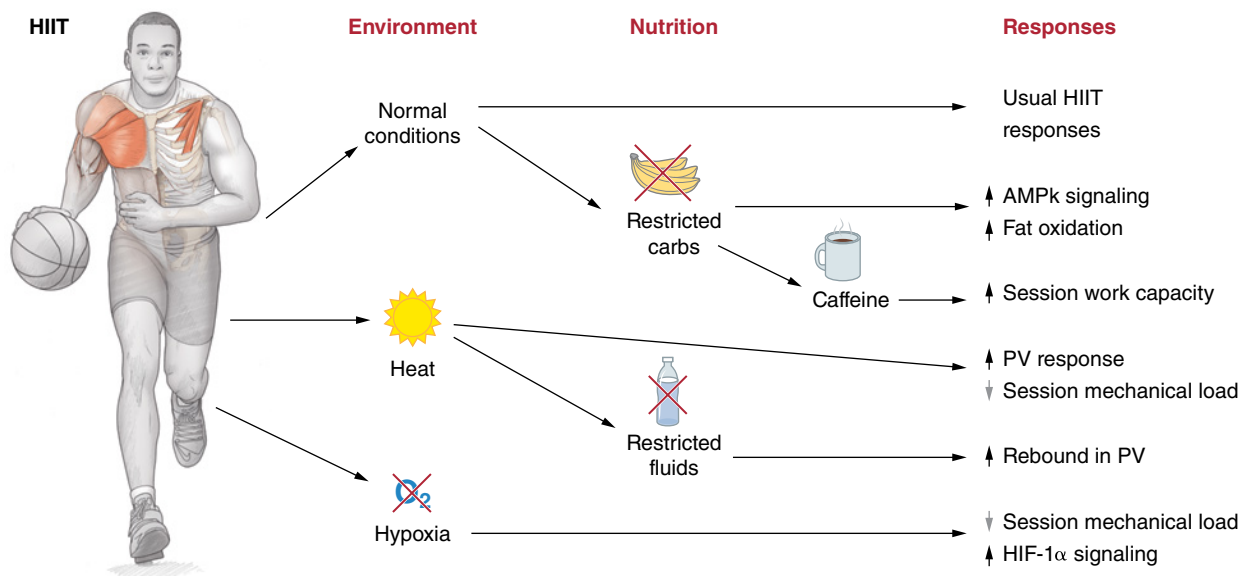


Figure 4.12 Impact of environment and nutrition and their associated acute responses.

by short durations (1 to 3 min) of passive recovery or longer durations of active recovery up to 45% of V_{IFT} or 60% of $V/P_{IncTest}$ (2 to 4 min; figure 4.13).

Short Intervals

Short intervals (figure 4.14) consist of interval bouts of less than 60 s repeated in a similarly short time duration. The dynamics of this format enable us to induce the acute metabolic and neuromuscular responses we are after, which we will allude to in the next chapter. For short intervals to form effective weapons we can use to hit our targets, they should be performed at between 90% and 105% of V_{IFT} (100% to 120% $V/P_{IncTest}$) for their repeated short duration (10 to 60 s), separated by less than a minute of recovery (passive to 45% V_{IFT} or 60% $V/P_{IncTest}$).

Repeated-Sprint Training

Repeated-sprint training is the first of our very high-intensity weapon formats we can use to target more high-end capacities or anaerobic speed/power reserve. These formats (figure 4.15) involve 3 to 10 s efforts of an all-out maximal intensity, with a variable recovery duration ranging from short and passive, to 45% V_{IFT} or 60% $V/P_{IncTest}$.

Sprint Interval Training

Sprint interval training is again all-out maximal sprinting efforts, only the duration is longer, in the 20 to 45 s range. Such efforts are exceptionally taxing, and recovery is passive and long (typically 1 to 4 min). Sprint interval training weapon formats are shown in figure 4.16.

Game-Based HIIT

Game-based HIIT (GBHIIT), including small-sided games (SSGs), discussed briefly in chapter 2 and expanded in detail in chapter 5, are ultimately sport-specific game-based forms of long intervals. Importantly, GBHIIT systematically includes decision-making and interactions with opponents (at least one in the case of racket sports) and teammates, which make these unique and different from typical sport-specific intervals (e.g., a tennis player hitting successive, but different, balls during a short interval). GBHIIT tends to run for 2 to 4 min at a sport-specific intensity of effort, which outside of the research setting can be difficult to quantify. Recovery durations are typically passive and range from 90 s to 4 min. Figure 4.17 illustrates a schematic representation of this weapon format.

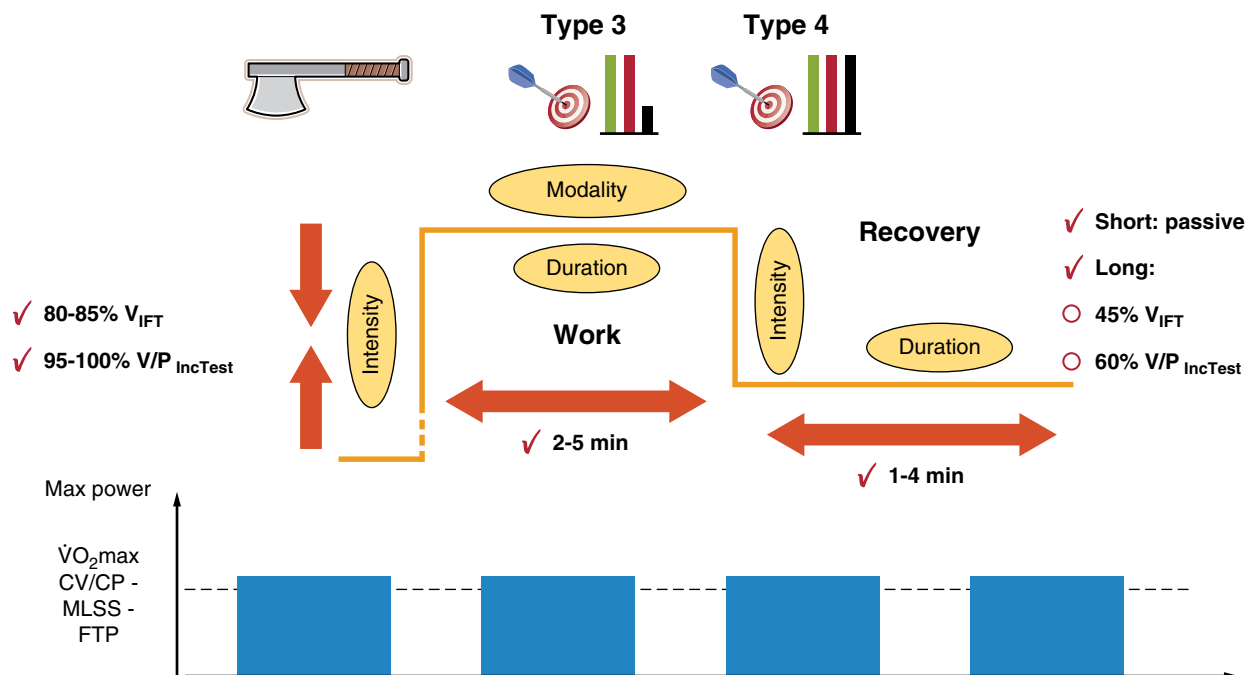


Figure 4.13 Long intervals and their recommended range of intensities, durations, and recovery characteristics.

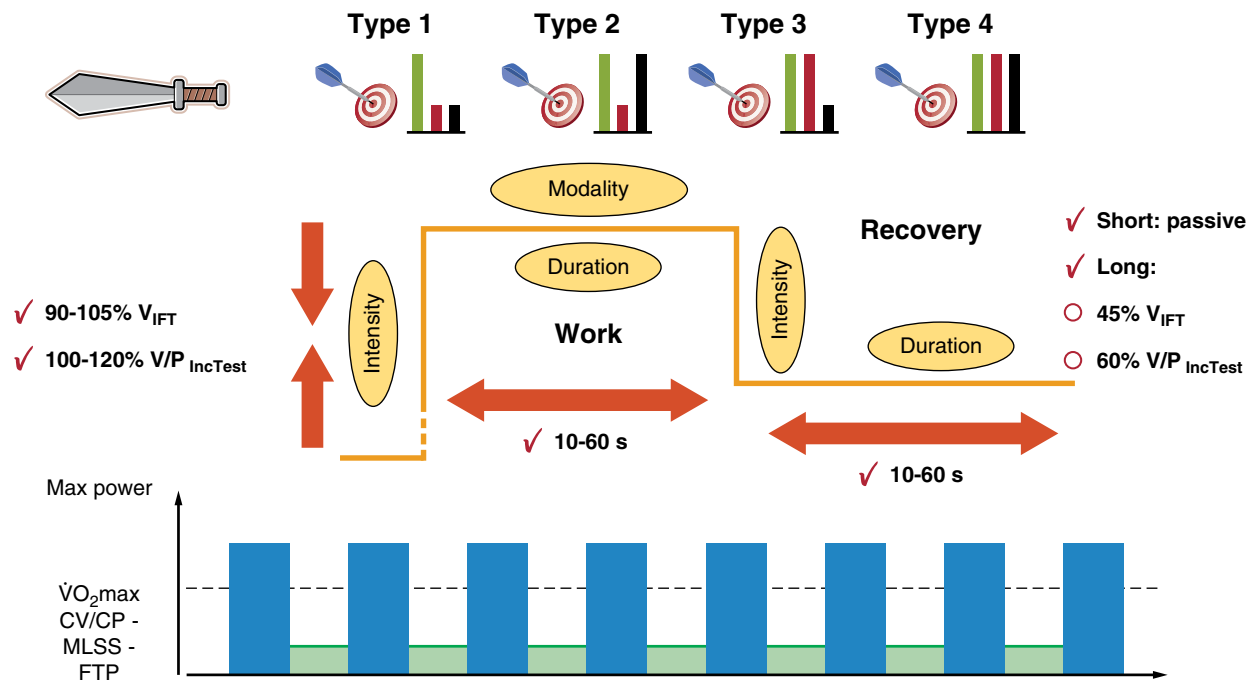


Figure 4.14 Short intervals and their recommended range of intensities, durations, and recovery characteristics.

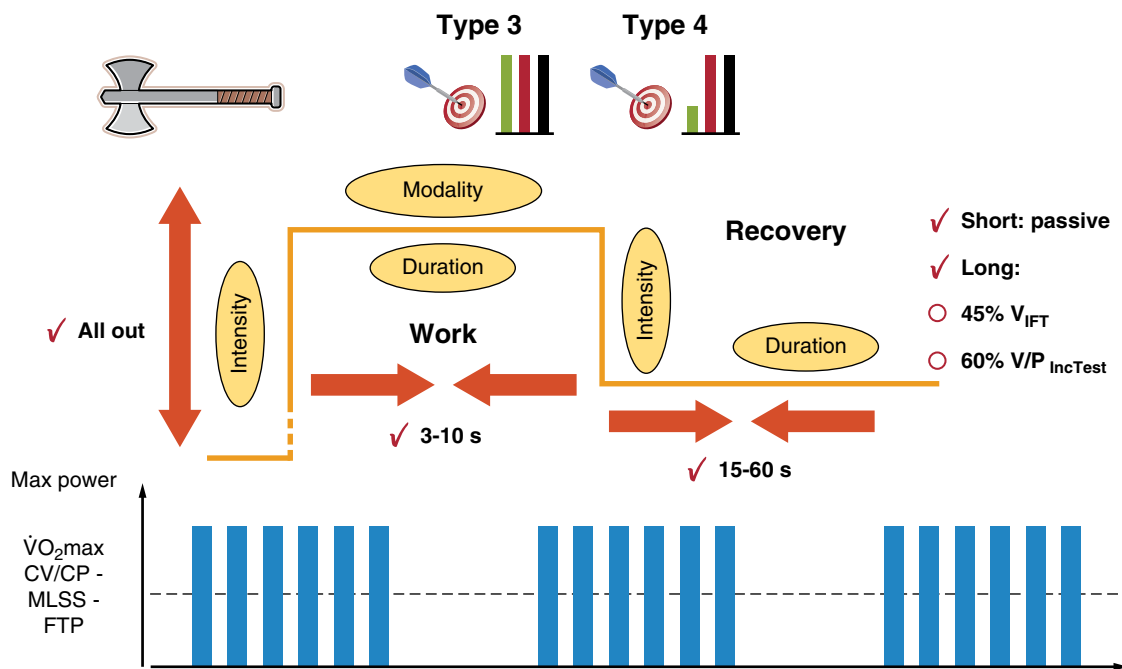


Figure 4.15 Repeated-sprint training and its recommended range of intensities, durations, and recovery characteristics.

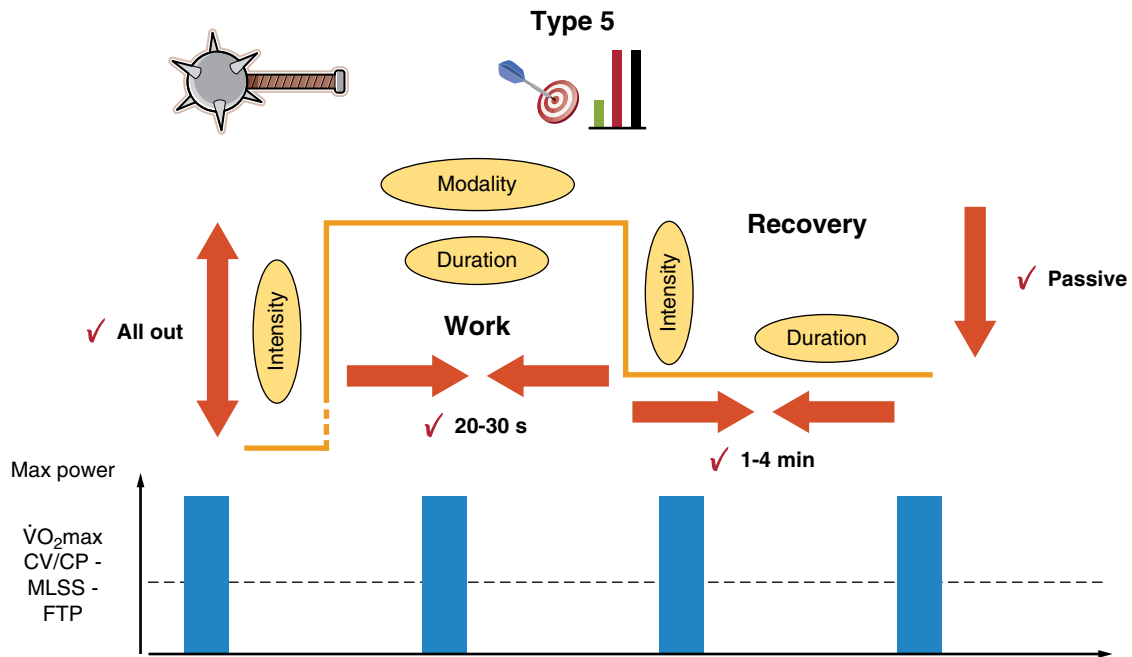


Figure 4.16 Sprint interval training.

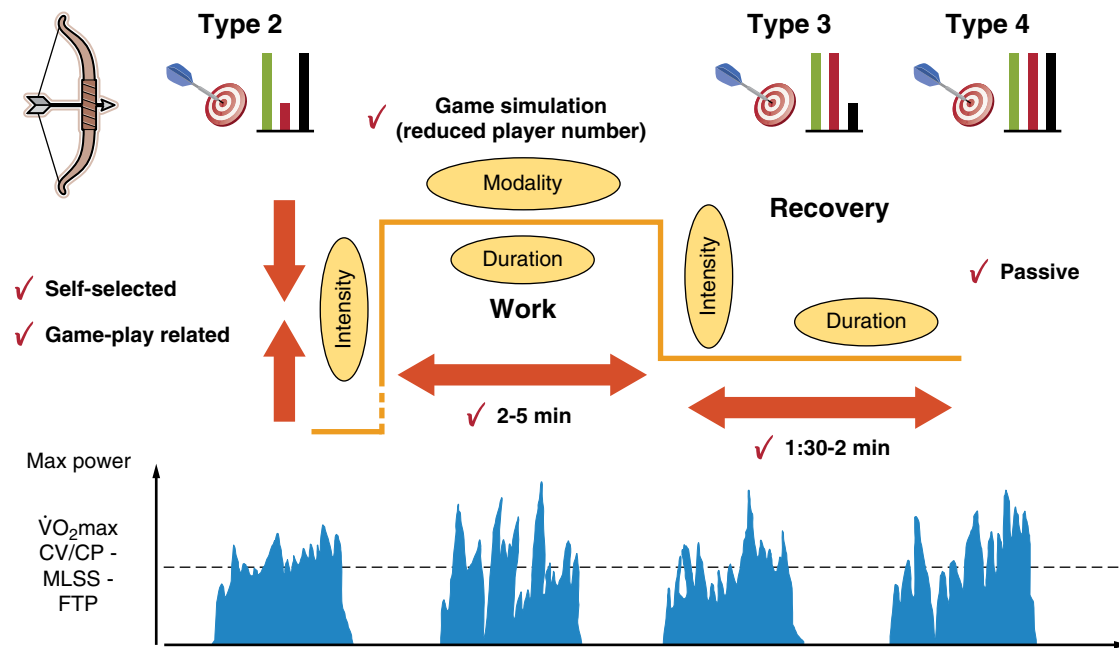


Figure 4.17 Game-based HIIT.

Summary

In summary, we have shown the 12 major factors we can manipulate when it comes to the development of an HIIT weapon. These were shown to be (1) the intensity of the work bout; (2) the duration of the work bout; (3) the intensity of the recovery period; (4) the duration of the recovery period; (5) the number of intervals or series duration; (6) the number of interval bout series; (7) the between-series recovery duration and (8) intensity; and, finally, (9) the total work performed. The other factors that

play a large role on the physiological outcome of an HIIT session include (10) the exercise mode and ground surface for run-based HIIT; (11) the environment (heat and altitude); and (12) an athlete's nutrition practices shortly before (<24 h) and during the session.

While the foundations of our five key weapons are now formed, we need to go deeper with this information and further fine-tune these. Thus, in chapter 5 we will use our knowledge of how to skin a cat to refine and form more elaborate weapons that we can use to hit our military targets.